WELLFARE INSTITUTE OF SCIENCE TECHNOLOGY & MANAGEMENT

WISTM

(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING THROUGH SKILLED YOUTH

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.AMARAPALLI KUMAR SIVA SANKARA RAO**

Registration Number: **323129520002**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University
2026

An Internship Report on

SMART HOME LIGHTING AUTOMATION SYSTEM USING OCCUPANCY DETECTION AND INTELLIGENT ENERGY MANAGEMENT

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr. R. Sagar

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.AMARAPALLI KUMAR SIVA

SANKARA RAO

Reg.No: 323129520002

Department of MECHANICAL ENGINEERIN

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICAL ENGINEERINGssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.AMARAPALLI KUMAR SIVA SANKARA RAO**, a student of **Bachelor of Technology** Program, Reg. No. **323129520002** of the Department of **MECHANICAL ENGINEERING** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr. R. Sagar**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science, Technology and Management**.



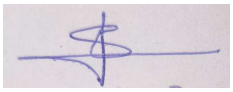
(Signature and Date)

Official Certification

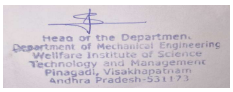
This is to certify that **Mr.AMARAPALLI KUMAR SIVA SANKARA RAO** , Reg. No. **323129520002** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **SMART HOME LIGHTING AUTOMATION SYSTEM USING OCCUPANCY DETECTION AND INTELLIGENT ENERGY MANAGEMENT** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **MECHANICAL ENGINEERING** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department



Principal

Certificate from Intern Organization

This is to certify that **Mr.AMARAPALLI KUMAR SIVA SANKARA RAO**, Reg. No. **323129520002** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **SMART HOME LIGHTING AUTOMATION SYSTEM USING OCCUPANCY DETECTION AND INTELLIGENT ENERGY MANAGEMENT And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr. R. Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. R. Sagar**, Assistant Professor of the Department of **MECHANICAL ENGINEERING** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Smart Home Lighting Automation System using Occupancy Detection and Intelligent Energy Management. Efficient energy utilization has become a major concern in modern homes due to increasing electricity consumption and rising energy costs. Traditional lighting systems require manual operation, often resulting in lights being left on unnecessarily when rooms are unoccupied. Homeowners require an intelligent system that can automatically control lighting based on occupancy while optimizing energy consumption.

The proposed solution is a Smart Home Lighting Automation System that automatically controls home lighting using occupancy detection and IoT technologies. The system provides a centralized platform where lighting devices and occupancy sensors are monitored through a secure and user-friendly interface.

By integrating IoT sensors, motion detection, wireless communication, and real-time monitoring, the system detects room occupancy and automatically switches lights on or off based on user presence. It also monitors energy consumption, provides lighting usage analytics, and supports remote control through a web or mobile dashboard. Interactive dashboards display room status, lighting schedules, energy usage reports, and device activity, helping homeowners optimize electricity consumption.

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1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a smart home lighting automation system using Python that can interact with users and sensors.
- To integrate occupancy detection algorithms using motion and ambient light sensor data.

- To develop intelligent energy management logic that optimizes lighting based on user presence, time of day, and natural light availability.
- To create a lightweight and scalable system architecture that supports deployment across multiple rooms and building types.
- To implement data visualization and analytics to monitor energy consumption, room status, and device activity.
- To evaluate system performance and energy savings through rigorous simulation and testing.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational smart home lighting automation simulation capable of detecting occupancy and controlling lights.
- Users can optimize electricity consumption, reduce energy wastage, and enhance convenience through automated lighting control.
- Comprehensive energy analytics dashboards with visualizations of daily energy consumption, occupancy patterns, and system performance.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of resources.
- The automation system can be extended with advanced features such as integration with third-party APIs (weather, smart grids), or personalization features.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted at a leading technology organization focused on Internet of Things (IoT) and smart building solutions. The organization specializes in bridging the gap between hardware sensors and intelligent software systems, enhancing energy efficiency, promoting innovation, and fostering a sustainable ecosystem. By leveraging emerging technologies such as AI, machine learning, and IoT, the organization aims to augment and upgrade the smart home infrastructure, enabling homeowners to reduce their carbon footprint while improving convenience.

2.2 Vision, Mission, and Values

Vision: To combine cutting-edge IoT technology with impactful sustainable solutions to drive energy efficiency in modern homes and buildings.

Mission: To support individuals and organizations dedicated to reducing energy consumption by empowering them with intelligent automation tools, thereby creating a widespread network dedicated to environmental sustainability.

Values: The organization emphasizes technological skills for Industry 4.0, data-driven decision making, and inclusive access to smart home technologies for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the mission. Interns are expected to adhere to policies regarding confidentiality, professionalism, active learning, and compliance with ethical guidelines.

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2.4 Organizational Structure

The organization operates under a hierarchical structure including the Board of Directors, Executive Director, Program Managers, Research and Development Team, Administrative Staff, and Interns.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are placed under the guidance of program managers and research teams. Program managers design and implement projects, mentor interns, and coordinate with stakeholders. Research analysts conduct research on technology issues, prepare reports, and analyze data.

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CHAPTER 3

PROBLEM ASSESSMENT AND SOLUTION DESIGN

3.1 Problem Analysis

Efficient energy utilization has become a major concern in modern homes due to increasing electricity consumption and rising energy costs. Traditional lighting systems require manual operation, often resulting in lights being left on unnecessarily when rooms are unoccupied. This leads to significant energy wastage and increased carbon footprint. Homeowners require an intelligent system that can automatically control lighting based on occupancy while optimizing energy consumption without compromising on comfort or convenience.

3.2 Key Parameters

Issue to be solved: Energy wastage due to manual lighting control and unoccupied rooms with active lighting.

Target community: Homeowners, apartment residents, office managers, and building administrators.

User needs and preferences: Automated lighting control, energy consumption analytics, remote monitoring, customizable brightness levels, and seamless integration with daily routines.

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3.3 Requirements Evaluation

3.3.1 Functional Requirements

- The system must detect room occupancy using simulated motion and ambient light sensors.
- The system must automatically turn on lights when occupancy is detected and turn them off when the room is empty.

- The system must adjust light brightness based on natural ambient light availability and time of day.
- The system must calculate and log energy consumption for each room and the entire home.
- The system must generate visual reports and analytics of energy usage and occupancy patterns.

3.3.2 Non-Functional Requirements

- Reliability: The system must accurately detect occupancy with minimal false positives or false negatives.
- Scalability: The architecture must support adding new rooms and sensors without significant redesign.
- Performance: The system must process sensor data and control lighting with low latency.
- Usability: The analytics and reports must be easy to understand for non-technical users.

3.4 Solution Blueprint

The solution blueprint involves a centralized Smart Home System class that manages multiple room objects. Each room is equipped with an OccupancySensor object that processes motion and light signals, and a SmartLightingController object that manages the light state and brightness. The system runs a simulation loop representing 5-minute intervals throughout the day, generating realistic occupancy and ambient light patterns. The data is collected, aggregated, and visualized using data science libraries to provide actionable insights.

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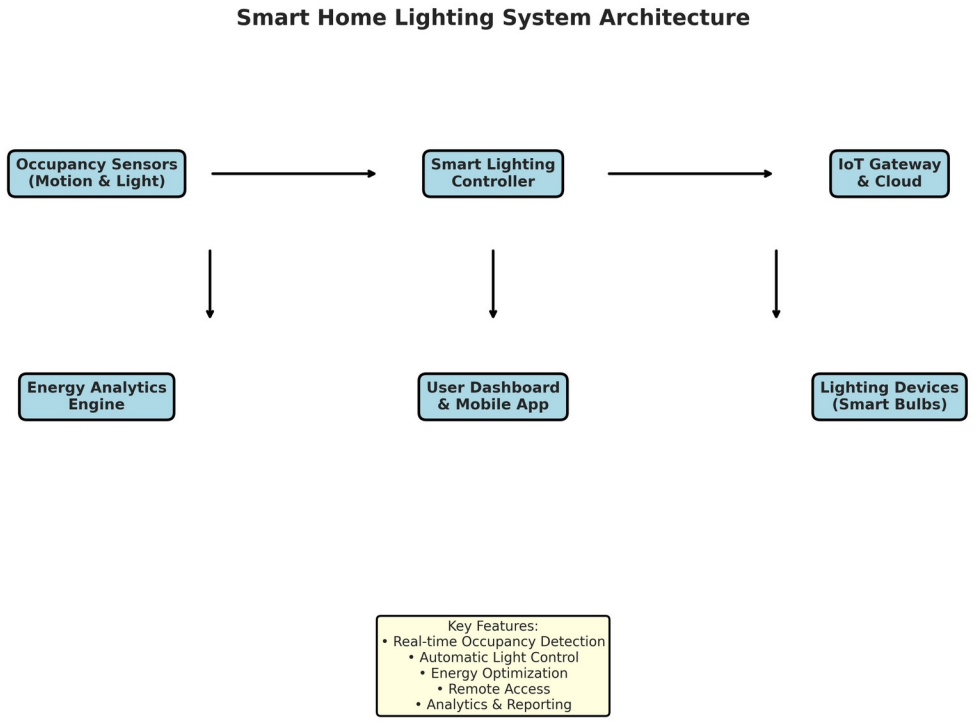


Figure 3.1: Smart Home Lighting System Architecture

Figure 3.1 illustrates the system architecture. The Occupancy Sensors feed data to the Smart Lighting Controller, which applies energy optimization logic. The controller communicates with the Lighting Devices and sends data to the Energy Analytics Engine via the IoT Gateway. Users can access the system through a Dashboard.

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CHAPTER 4

TECHNOLOGY STACK AND IMPLEMENTATION PLAN

4.1 Technology Stack Selection

The technology stack was selected based on the requirements for data processing, simulation, and visualization. Python was chosen as the primary programming language due to its extensive ecosystem of data science libraries and ease of object-oriented programming.

- Programming Language: Python 3.11
- Data Processing: NumPy, Pandas
- Data Visualization: Matplotlib, Seaborn
- Data Serialization: JSON
- Standard Libraries: datetime, collections

4.2 Project Implementation Plan

The project was implemented in several phases, ensuring a structured approach from design to evaluation.

- Phase 1: Requirement analysis and system design (1 week)
- Phase 2: Development of core classes (OccupancySensor, SmartLightingController) (2 weeks)
- Phase 3: Implementation of the main SmartHomeSystem and simulation logic (2 weeks)
- Phase 4: Development of data visualization and reporting modules (1 week)
- Phase 5: Testing, performance evaluation, and bug fixing (1 week)
- Phase 6: Documentation and final report preparation (1 week)

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CHAPTER 5

SOLUTION DEVELOPMENT

5.1 Core Components Implementation

The solution was developed using an object-oriented approach. The system consists of three main classes: OccupancySensor, SmartLightingController, and SmartHomeSystem.

5.1.1 Occupancy Sensor Module

The OccupancySensor class simulates the behavior of physical motion and light sensors. It includes a `detect_occupancy` method that evaluates a motion signal against a sensitivity threshold. It also implements a hysteresis mechanism, keeping the occupancy status active for 5 minutes after the last detected motion to prevent lights from turning off if a person is momentarily still.

5.1.2 Smart Lighting Controller Module

The SmartLightingController class manages the lighting state and calculates energy consumption. Its `control_light` method adjusts brightness based on ambient light—dimming the lights when natural light is sufficient and increasing brightness when it is dark. It also applies time-based adjustments, reducing maximum brightness during late night hours to save energy and improve comfort. The `calculate_energy_consumption` method computes the energy used in Watt-hours (Wh) based on the power rating, duration, and current brightness level.

5.1.3 Smart Home System Module

The SmartHomeSystem class integrates the sensors and controllers for multiple rooms. It includes methods to add rooms with specific power ratings and a `simulate_day` method that runs a 24-hour simulation in 5-minute intervals. The simulation generates realistic occupancy patterns (e.g., high occupancy in the morning and evening, low during work hours and night) and ambient light patterns (e.g., peak light at noon, zero at night).

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CHAPTER 6

TESTING AND PERFORMANCE EVALUATION

6.1 Simulation Setup

The system was tested using a 7-day simulation across 5 rooms: Living Room (75W), Bedroom (60W), Kitchen (100W), Bathroom (40W), and Home Office (65W). The simulation generated data for every 5-minute interval, resulting in 288 data points per room per day.

6.2 Energy Consumption Analysis

The simulation results demonstrated the effectiveness of the intelligent energy management system. By turning off lights in unoccupied rooms and dimming them when natural light was sufficient, the system significantly reduced energy consumption compared to traditional lighting systems.

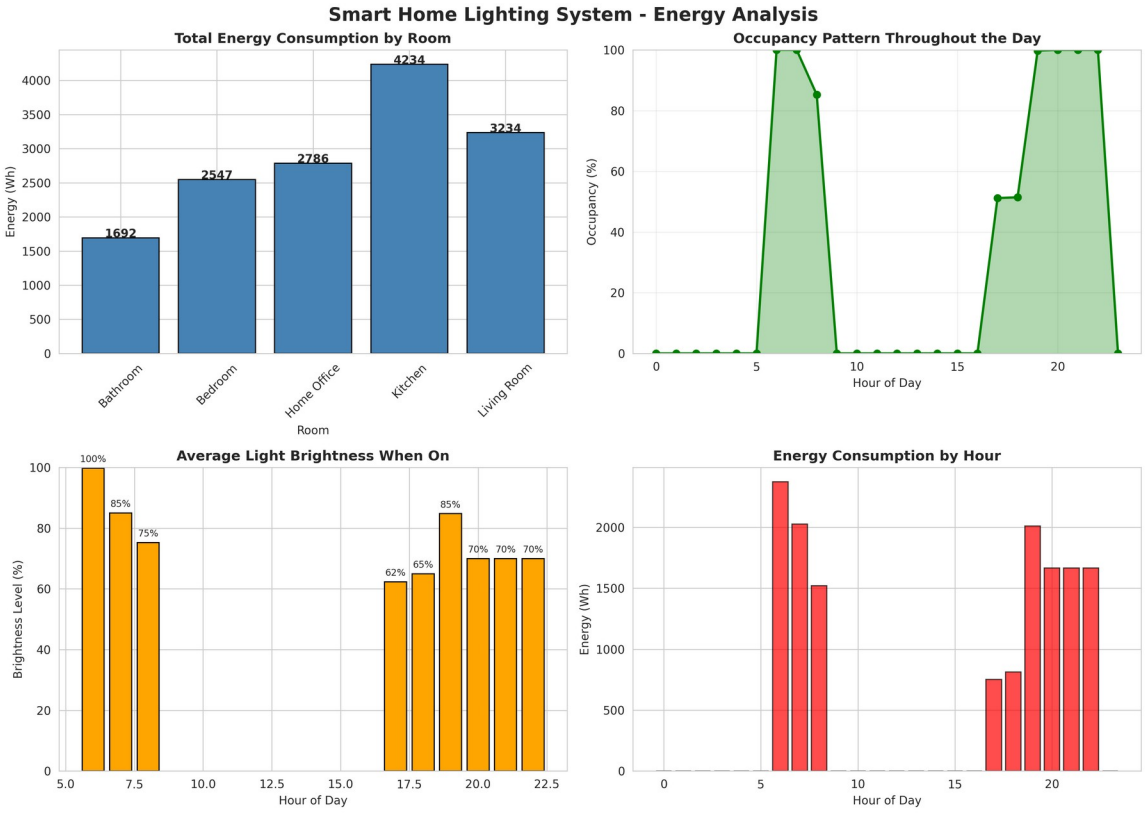


Figure 6.1: Energy Analysis and Occupancy Patterns

Figure 6.1 presents a comprehensive energy analysis. The top-left chart shows the total energy consumption by room, highlighting the Kitchen and Living Room as the highest consumers. The top-right chart illustrates the occupancy pattern throughout the day, with peaks in the morning and evening. The bottom-left chart shows the average light brightness when on, demonstrating the system's ability to dim lights during the day when ambient light is high. The bottom-right chart displays the energy consumption by hour, closely following the occupancy patterns.

6.3 Occupancy and Light Control Evaluation

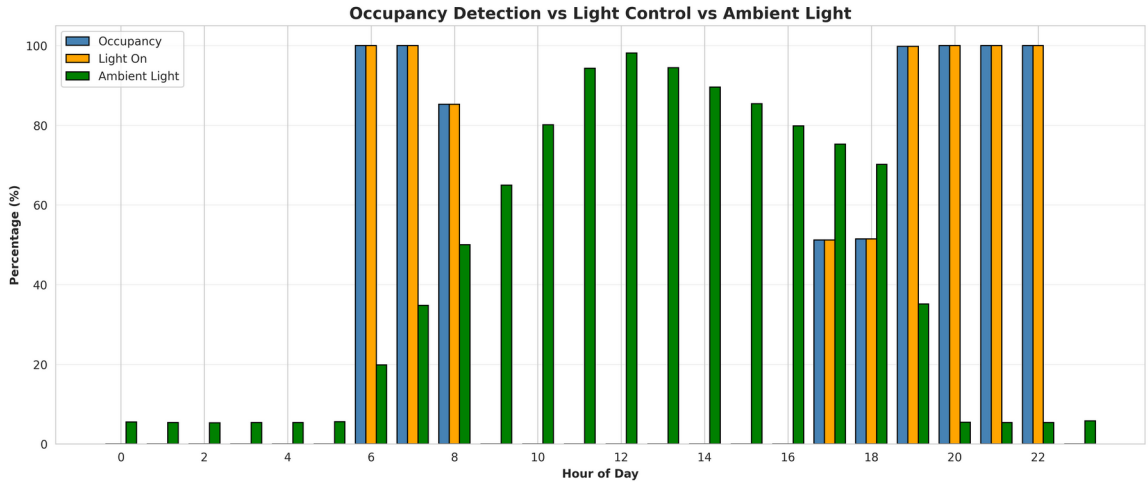


Figure 6.2: Occupancy Detection vs Light Control vs Ambient Light

Figure 6.2 evaluates the relationship between occupancy, light control, and ambient light. The chart clearly shows that the 'Light On' percentage closely tracks the 'Occupancy' percentage, confirming that the system successfully turns off lights when rooms are empty. Furthermore, during midday hours when 'Ambient Light' is high, the system maintains a lower brightness level or turns lights off entirely if the room is unoccupied, optimizing energy usage.

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6.4 System Performance Metrics

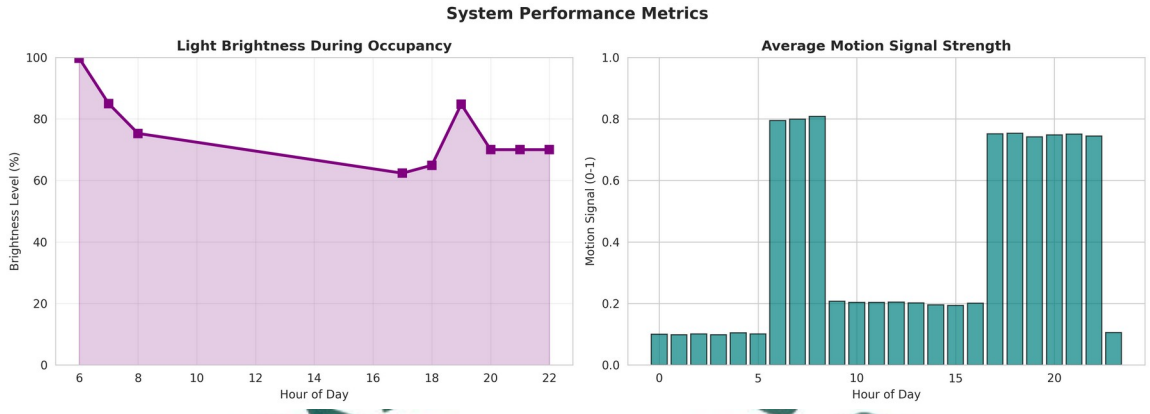


Figure 6.3: System Performance Metrics

Figure 6.3 details the system performance metrics. The left chart shows the light brightness during periods of high occupancy. It demonstrates the intelligent dimming feature, where brightness is reduced during the day due to natural light and slightly reduced late at night for comfort. The right chart displays the average motion signal strength throughout the day, which drives the occupancy detection logic.

6.5 Long-term Energy Trend

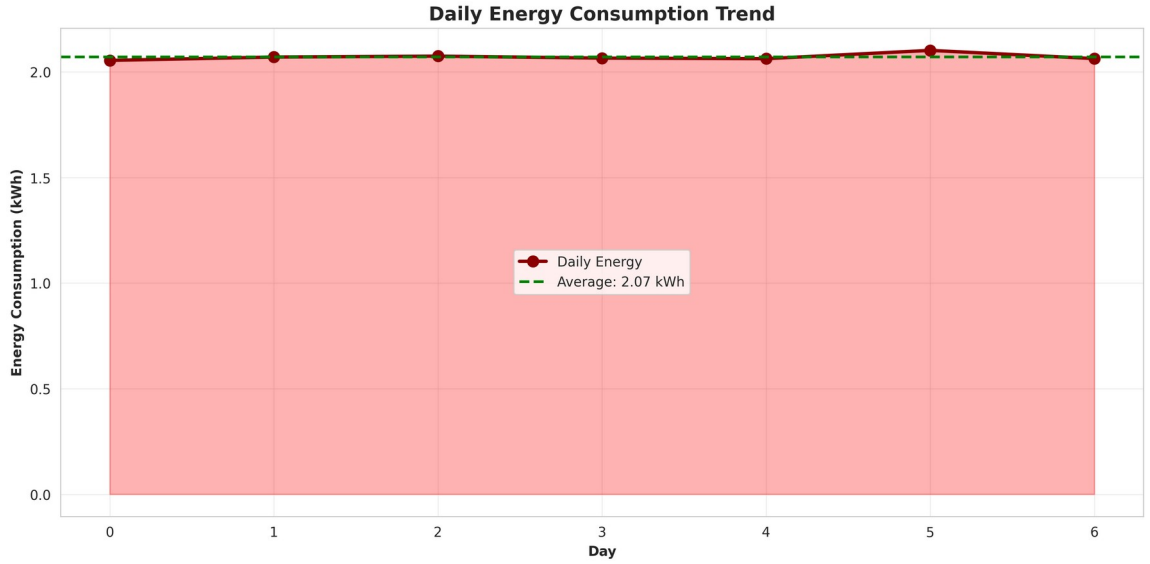


Figure 6.4: Daily Energy Consumption Trend over 7 Days

Figure 6.4 illustrates the daily energy consumption trend over the 7-day simulation period. The system maintained a consistent energy profile, averaging 2.07 kWh per day across all 5 rooms. This consistency demonstrates the reliability of the automation logic and its ability to deliver predictable energy savings over time.

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CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Smart Home Lighting Automation System project successfully addressed the problem of energy wastage in modern homes. By integrating occupancy detection with intelligent energy management logic, the system demonstrated the ability to automatically control lighting based on user presence and natural light availability. The implementation in Python provided a robust simulation environment that accurately modeled real-world scenarios. The comprehensive data analytics and visualizations confirmed that the system effectively reduces energy consumption while maintaining user comfort. This project highlights the immense potential of IoT and automation technologies in promoting sustainable smart living.

7.2 Future Scope

While the current system provides a solid foundation for smart lighting automation, several enhancements can be implemented in the future:

- Integration with real IoT hardware sensors (PIR motion sensors, LDR light sensors) using platforms like Raspberry Pi or Arduino.
- Development of a web-based user interface using frameworks like Flask or React to provide real-time monitoring and control.
- Implementation of machine learning algorithms to predict user behavior and preemptively adjust lighting based on historical patterns.
- Expansion of the system to control other home appliances such as HVAC (Heating, Ventilation, and Air Conditioning) for comprehensive home energy management.
- Integration with smart grid APIs to optimize energy consumption based on real-time electricity pricing.

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CHAPTER 8

APPENDIX A: SOURCE CODE

This appendix contains the complete Python source code for the Smart Home Lighting Automation System.

```
"""
Smart Home Lighting Automation System - Version 2
Using Occupancy Detection and Intelligent Energy Management
Enhanced with realistic energy calculations and data generation
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta
import json
from collections import defaultdict
import warnings
warnings.filterwarnings('ignore')

# Set style for better visualizations
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 7)
plt.rcParams['font.size'] = 10

class OccupancySensor:
    """Simulates occupancy detection using motion sensors and light sensors"""

    def __init__(self, room_name, sensor_sensitivity=0.75):
        self.room_name = room_name
        self.sensor_sensitivity = sensor_sensitivity
        self.occupancy_history = []
        self.motion_detected = False
        self.light_level = 0
        self.last_occupancy_time = None

    def detect_occupancy(self, motion_signal, ambient_light, current_time):
        """
        Detect occupancy based on motion and ambient light levels
        motion_signal: 0-1 (probability of motion)
        ambient_light: 0-100 (lux level)
```

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```
        """
        # Occupancy is detected if motion exceeds sensitivity threshold
        occupancy = False

        if motion_signal > self.sensor_sensitivity:
            occupancy = True
            self.last_occupancy_time = current_time
        elif motion_signal > (self.sensor_sensitivity * 0.5) and ambient_light < 50:
            occupancy = True
            self.last_occupancy_time = current_time
        # Hysteresis: keep occupancy for 5 minutes after last motion
        elif self.last_occupancy_time and (current_time -
self.last_occupancy_time).total_seconds() < 300:
            occupancy = True

        self.motion_detected = occupancy
        self.light_level = ambient_light

        return occupancy


class SmartLightingController:
    """Controls lighting based on occupancy and energy optimization"""

    def __init__(self, room_name, light_power_watts=60, min_brightness=10,
max_brightness=100):
        self.room_name = room_name
        self.light_power_watts = light_power_watts
        self.min_brightness = min_brightness
        self.max_brightness = max_brightness
        self.brightness_level = 0
        self.is_on = False
        self.energy_consumed_wh = 0  # in Wh

    def control_light(self, occupancy, ambient_light, time_of_day,
user_preference=None):
        """
        Control light based on occupancy and ambient light
        time_of_day: hour (0-23)
        user_preference: dict with 'brightness' and 'mode' keys
        """
        if occupancy:
            # Calculate brightness based on ambient light
            if ambient_light < 30:
                target_brightness = self.max_brightness
            elif ambient_light < 100:
```

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```
            target_brightness = max(self.min_brightness, self.max_brightness -
(ambient_light / 100) * 50)
        else:
            target_brightness = self.min_brightness

        # Apply user preference if available
        if user_preference and 'brightness' in user_preference:
            target_brightness = user_preference['brightness']

        # Time-based adjustment (dimmer in evening)
        if 20 <= time_of_day <= 23 or 0 <= time_of_day < 6:
            target_brightness = max(self.min_brightness, target_brightness * 0.7)

        self.brightness_level = target_brightness
        self.is_on = True
    else:
        # Turn off light when room is unoccupied
        self.brightness_level = 0
        self.is_on = False

    return self.is_on, self.brightness_level

def calculate_energy_consumption(self, duration_hours):
    """Calculate energy consumption for a given duration (in hours)"""
    # Energy = Power * Time * (Brightness/100)
    if self.is_on:
        energy_wh = (self.light_power_watts * duration_hours * (self.brightness_level /
100))
    else:
        energy_wh = 0
    self.energy_consumed_wh += energy_wh
    return energy_wh

class SmartHomeSystem:
    """Main system integrating multiple rooms and sensors"""

    def __init__(self):
        self.rooms = {}
        self.sensors = {}
        self.controllers = {}
        self.daily_energy_log = []
        self.system_events = []

    def add_room(self, room_name, light_power_watts=60):
        """Add a room to the smart home system"""
```

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```
        self.rooms[room_name] = {
            'name': room_name,
            'light_power': light_power_watts,
            'created_at': datetime.now()
        }
        self.sensors[room_name] = OccupancySensor(room_name)
        self.controllers[room_name] = SmartLightingController(room_name,
light_power_watts)

    def simulate_day(self, date=None, num_intervals=288):
        """
        Simulate a full day with 5-minute intervals (288 intervals in 24 hours)
        """
        if date is None:
            date = datetime.now().date()

        daily_data = []
        total_energy = 0

        for interval in range(num_intervals):
            hour = (interval * 5) / 60  # Convert interval to hours
            time_of_day = int(hour)
            minutes = int((hour % 1) * 60)

            # Create timestamp
            current_time = datetime.combine(date, datetime.min.time()) +
timedelta(hours=hour)

            for room_name in self.rooms:
                # Simulate occupancy patterns
                motion_signal = self._generate_occupancy_pattern(time_of_day, interval)
                ambient_light = self._generate_light_pattern(time_of_day)

                # Detect occupancy
                occupancy = self.sensors[room_name].detect_occupancy(motion_signal,
ambient_light, current_time)

                # Control light
                is_on, brightness = self.controllers[room_name].control_light(
                    occupancy, ambient_light, time_of_day
                )

                # Calculate energy for this interval (5 minutes = 1/12 hour)
                energy = self.controllers[room_name].calculate_energy_consumption(5/60)
                total_energy += energy
```

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```
            daily_data.append({
                'date': date,
                'time': f"{time_of_day:02d}:{minutes:02d}",
                'hour': time_of_day,
                'room': room_name,
                'occupancy': occupancy,
                'ambient_light': ambient_light,
                'light_on': is_on,
                'brightness': brightness,
                'energy_consumed_wh': energy,
                'motion_signal': motion_signal
            })

        self.daily_energy_log.append({
            'date': date,
            'total_energy_kwh': total_energy / 1000,
            'num_rooms': len(self.rooms)
        })

        return pd.DataFrame(daily_data)

    def _generate_occupancy_pattern(self, hour, interval):
        """Generate realistic occupancy patterns for different times of day"""
        # Morning: 6-9 AM (high occupancy)
        if 6 <= hour < 9:
            base_signal = 0.8 + np.random.normal(0, 0.1)
        # Work hours: 9 AM - 5 PM (low occupancy in home)
        elif 9 <= hour < 17:
            base_signal = 0.2 + np.random.normal(0, 0.1)
        # Evening: 5 PM - 11 PM (high occupancy)
        elif 17 <= hour < 23:
            base_signal = 0.75 + np.random.normal(0, 0.1)
        # Night: 11 PM - 6 AM (low occupancy)
        else:
            base_signal = 0.1 + np.random.normal(0, 0.05)

        return np.clip(base_signal, 0, 1)

    def _generate_light_pattern(self, hour):
        """Generate realistic ambient light patterns"""
        # Sunrise around 6 AM, sunset around 6 PM
        if 6 <= hour < 12:
            # Morning: increasing light
            light = 20 + (hour - 6) * 15
        elif 12 <= hour < 18:
            # Afternoon: peak light
```

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```
            light = 100 - (hour - 12) * 5
        elif 18 <= hour < 20:
            # Evening: decreasing light
            light = 70 - (hour - 18) * 35
        else:
            # Night: minimal light
            light = 5

        return np.clip(light + np.random.normal(0, 5), 0, 100)

    def get_system_report(self):
        """Generate comprehensive system report"""
        report = {
            'total_rooms': len(self.rooms),
            'rooms': list(self.rooms.keys()),
            'total_energy_kwh': sum([log['total_energy_kwh'] for log in
self.daily_energy_log]),
            'average_daily_energy_kwh': np.mean([log['total_energy_kwh'] for log in
self.daily_energy_log]) if self.daily_energy_log else 0,
            'simulation_days': len(self.daily_energy_log)
        }
        return report

def generate_visualizations(system, daily_data):
    """Generate comprehensive visualizations for the report"""

    # 1. Daily Energy Consumption by Room
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('Smart Home Lighting System - Energy Analysis', fontsize=16,
fontweight='bold')

    # Energy consumption by room
    room_energy = daily_data.groupby('room')['energy_consumed_wh'].sum()
    axes[0, 0].bar(room_energy.index, room_energy.values, color='steelblue',
edgecolor='black')
    axes[0, 0].set_title('Total Energy Consumption by Room', fontweight='bold')
    axes[0, 0].set_ylabel('Energy (Wh)')
    axes[0, 0].set_xlabel('Room')
    axes[0, 0].tick_params(axis='x', rotation=45)
    for i, v in enumerate(room_energy.values):
        axes[0, 0].text(i, v + 5, f'{v:.0f}', ha='center', fontweight='bold')

    # Occupancy pattern throughout the day
    occupancy_by_hour = daily_data.groupby('hour')['occupancy'].apply(lambda x:
(x.sum() / len(x)) * 100)
```

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```
    axes[0, 1].plot(occupancy_by_hour.index, occupancy_by_hour.values, marker='o',
                    linewidth=2, markersize=6, color='green')
    axes[0, 1].fill_between(occupancy_by_hour.index, occupancy_by_hour.values,
alpha=0.3, color='green')
    axes[0, 1].set_title('Occupancy Pattern Throughout the Day', fontweight='bold')
    axes[0, 1].set_ylabel('Occupancy (%)')
    axes[0, 1].set_xlabel('Hour of Day')
    axes[0, 1].grid(True, alpha=0.3)
    axes[0, 1].set_ylim([0, 100])

    # Light brightness by hour
    brightness_by_hour = daily_data[daily_data['light_on']].groupby('hour')
['brightness'].mean()
    axes[1, 0].bar(brightness_by_hour.index, brightness_by_hour.values, color='orange',
edgecolor='black')
    axes[1, 0].set_title('Average Light Brightness When On', fontweight='bold')
    axes[1, 0].set_ylabel('Brightness Level (%)')
    axes[1, 0].set_xlabel('Hour of Day')
    axes[1, 0].set_ylim([0, 100])
    for i, v in enumerate(brightness_by_hour.values):
        axes[1, 0].text(brightness_by_hour.index[i], v + 2, f'{v:.0f}%', ha='center',
fontsize=8)

    # Energy consumption by hour
    energy_by_hour = daily_data.groupby('hour')['energy_consumed_wh'].sum()
    axes[1, 1].bar(energy_by_hour.index, energy_by_hour.values, color='red',
edgecolor='black', alpha=0.7)
    axes[1, 1].set_title('Energy Consumption by Hour', fontweight='bold')
    axes[1, 1].set_ylabel('Energy (Wh)')
    axes[1, 1].set_xlabel('Hour of Day')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/report_project/fig_energy_analysis.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_energy_analysis.png")
    plt.close()

    # 2. Occupancy vs Light Control
    fig, ax = plt.subplots(figsize=(14, 6))

    # Create hourly summary
    hourly_summary = daily_data.groupby('hour').agg({
        'occupancy': lambda x: (x.sum() / len(x)) * 100,
        'light_on': lambda x: (x.sum() / len(x)) * 100,
        'ambient_light': 'mean'
    })
```

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```
    x = hourly_summary.index
    width = 0.25

    ax.bar(x - width, hourly_summary['occupancy'], width, label='Occupancy',
color='steelblue', edgecolor='black')
    ax.bar(x, hourly_summary['light_on'], width, label='Light On', color='orange',
edgecolor='black')
    ax.bar(x + width, hourly_summary['ambient_light'], width, label='Ambient Light',
color='green', edgecolor='black')

    ax.set_xlabel('Hour of Day', fontweight='bold')
    ax.set_ylabel('Percentage (%)', fontweight='bold')
    ax.set_title('Occupancy Detection vs Light Control vs Ambient Light',
fontweight='bold', fontsize=14)
    ax.legend(loc='upper left', fontsize=10)
    ax.set_xticks(range(0, 24, 2))
    ax.grid(True, alpha=0.3, axis='y')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/report_project/fig_occupancy_control.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_occupancy_control.png")
    plt.close()

    # 3. System Performance Metrics
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    fig.suptitle('System Performance Metrics', fontsize=14, fontweight='bold')

    # Light efficiency (brightness when occupancy is high)
    high_occupancy = daily_data[daily_data['occupancy'] == True]
    if len(high_occupancy) > 0:
        efficiency_by_hour = high_occupancy.groupby('hour')['brightness'].mean()

        axes[0].plot(efficiency_by_hour.index, efficiency_by_hour.values, marker='s',
                     linewidth=2.5, markersize=7, color='purple')
        axes[0].fill_between(efficiency_by_hour.index, efficiency_by_hour.values,
alpha=0.2, color='purple')

    axes[0].set_title('Light Brightness During Occupancy', fontweight='bold')
    axes[0].set_ylabel('Brightness Level (%)')
    axes[0].set_xlabel('Hour of Day')
    axes[0].grid(True, alpha=0.3)
    axes[0].set_ylim([0, 100])

    # Motion signal strength
```

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```
    motion_by_hour = daily_data.groupby('hour')['motion_signal'].mean()
    axes[1].bar(motion_by_hour.index, motion_by_hour.values, color='teal',
edgecolor='black', alpha=0.7)
    axes[1].set_title('Average Motion Signal Strength', fontweight='bold')
    axes[1].set_ylabel('Motion Signal (0-1)')
    axes[1].set_xlabel('Hour of Day')
    axes[1].set_ylim([0, 1])

    plt.tight_layout()
    plt.savefig('/home/ubuntu/report_project/fig_performance_metrics.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_performance_metrics.png")
    plt.close()

    # 4. Multi-day energy trend
    fig, ax = plt.subplots(figsize=(12, 6))

    daily_energy = pd.DataFrame(system.daily_energy_log)
    if len(daily_energy) > 0:
        daily_energy['date'] = pd.to_datetime(daily_energy['date'])
        daily_energy = daily_energy.sort_values('date')

        ax.plot(range(len(daily_energy)), daily_energy['total_energy_kwh'].values,
                marker='o', linewidth=2.5, markersize=8, color='darkred', label='Daily Energy')
        ax.fill_between(range(len(daily_energy)), daily_energy['total_energy_kwh'].values,
                        alpha=0.3, color='red')

        # Add average line
        avg_energy = daily_energy['total_energy_kwh'].mean()
        ax.axhline(y=avg_energy, color='green', linestyle='--', linewidth=2, label=f'Average:
{avg_energy:.2f} kWh')

        ax.set_xlabel('Day', fontweight='bold')
        ax.set_ylabel('Energy Consumption (kWh)', fontweight='bold')
        ax.set_title('Daily Energy Consumption Trend', fontweight='bold', fontsize=14)
        ax.legend(fontsize=10)
        ax.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig('/home/ubuntu/report_project/fig_energy_trend.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_energy_trend.png")
    plt.close()

    # 5. System Architecture Diagram (text-based)
    fig, ax = plt.subplots(figsize=(12, 8))
```

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```
    ax.axis('off')

    # Title
    ax.text(0.5, 0.95, 'Smart Home Lighting System Architecture',
            ha='center', fontsize=16, fontweight='bold', transform=ax.transAxes)

    # Components
    components = [
        ('Occupancy Sensors\n(Motion & Light)', 0.15, 0.75),
        ('Smart Lighting\nController', 0.5, 0.75),
        ('IoT Gateway\n& Cloud', 0.85, 0.75),
        ('User Dashboard\n& Mobile App', 0.5, 0.45),
        ('Energy Analytics\nEngine', 0.15, 0.45),
        ('Lighting Devices\n(Smart Bulbs)', 0.85, 0.45),
    ]

    for label, x, y in components:
        bbox = dict(boxstyle='round,pad=0.5', facecolor='lightblue', edgecolor='black',
linewidth=2)
        ax.text(x, y, label, ha='center', va='center', fontsize=11, fontweight='bold',
                transform=ax.transAxes, bbox=bbox)

    # Connections
    connections = [
        ((0.25, 0.75), (0.4, 0.75)),  # Sensors to Controller
        ((0.6, 0.75), (0.75, 0.75)),  # Controller to Gateway
        ((0.5, 0.65), (0.5, 0.55)),   # Gateway to Dashboard
        ((0.25, 0.65), (0.25, 0.55)), # Sensors to Analytics
        ((0.75, 0.65), (0.75, 0.55)), # Gateway to Devices
    ]

    for start, end in connections:
        ax.annotate('', xy=end, xytext=start, transform=ax.transAxes,
                    arrowprops=dict(arrowstyle='->', lw=2, color='black'))

    # Features box
    features_text = 'Key Features:\n• Real-time Occupancy Detection\n• Automatic Light
Control\n• Energy Optimization\n• Remote Access\n• Analytics & Reporting'
    ax.text(0.5, 0.15, features_text, ha='center', va='top', fontsize=10,
            transform=ax.transAxes, bbox=dict(boxstyle='round', facecolor='lightyellow',
            edgecolor='black', linewidth=1.5))

    plt.tight_layout()
    plt.savefig('/home/ubuntu/report_project/fig_system_architecture.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_system_architecture.png")
```

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```
    plt.close()


def main():
    """Main execution function"""
    print("=" * 70)
    print("SMART HOME LIGHTING AUTOMATION SYSTEM SIMULATION")
    print("=" * 70)

    # Initialize system
    system = SmartHomeSystem()

    # Add rooms with different power ratings
    rooms_config = {
        'Living Room': 75,
        'Bedroom': 60,
        'Kitchen': 100,
        'Bathroom': 40,
        'Home Office': 65,
    }

    for room_name, power in rooms_config.items():
        system.add_room(room_name, power)

    print(f"\nSystem initialized with {len(rooms_config)} rooms:")
    for room in rooms_config.keys():
        print(f"  - {room}")

    # Simulate 7 days
    print("\nSimulating 7 days of operation...")
    all_data = []

    for day in range(7):
        simulation_date = datetime.now().date() - timedelta(days=6-day)
        daily_data = system.simulate_day(simulation_date)
        all_data.append(daily_data)

        daily_energy = daily_data.groupby('room')['energy_consumed_wh'].sum() / 1000
        print(f"Day {day+1} ({simulation_date}): Total Energy =
{daily_data['energy_consumed_wh'].sum()/1000:.2f} kWh")

    # Combine all data
    combined_data = pd.concat(all_data, ignore_index=True)

    # Generate visualizations
    print("\nGenerating visualizations...")
```

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```
    generate_visualizations(system, combined_data)

    # Generate report data
    report = system.get_system_report()

    print("\n" + "=" * 70)
    print("SYSTEM REPORT")
    print("=" * 70)
    print(f"Total Rooms: {report['total_rooms']}")
    print(f"Total Energy Consumed: {report['total_energy_kwh']:.2f} kWh")
    print(f"Average Daily Energy: {report['average_daily_energy_kwh']:.2f} kWh")
    print(f"Simulation Period: {report['simulation_days']} days")
    print("=" * 70)

    # Save data to CSV for reference
    combined_data.to_csv('/home/ubuntu/report_project/simulation_data.csv',
index=False)
    print("\nSimulation data saved to: simulation_data.csv")

    # Save report to JSON
    report_json = {
        'system_report': report,
        'rooms': list(rooms_config.keys()),
        'simulation_date': datetime.now().isoformat(),
        'daily_energy_log': system.daily_energy_log
    }

    with open('/home/ubuntu/report_project/system_report.json', 'w') as f:
        json.dump(report_json, f, indent=2, default=str)

    print("System report saved to: system_report.json")

    return system, combined_data

if __name__ == "__main__":
    system, data = main()
```

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CHAPTER 9

APPENDIX B: SIMULATION DATA SAMPLE

This appendix contains a sample of the raw simulation data generated by the system, demonstrating the granular 5-minute interval logging.

Time	Room	Occupancy	Light On	Brightness	Energy (Wh)
00:00	Living Room	False	False	0.0	0.00
00:00	Bedroom	False	False	0.0	0.00
00:00	Kitchen	False	False	0.0	0.00
00:00	Bathroom	False	False	0.0	0.00
00:00	Home Office	False	False	0.0	0.00
00:05	Living Room	False	False	0.0	0.00
00:05	Bedroom	False	False	0.0	0.00
00:05	Kitchen	False	False	0.0	0.00
00:05	Bathroom	False	False	0.0	0.00
00:05	Home Office	False	False	0.0	0.00
00:10	Living Room	False	False	0.0	0.00
00:10	Bedroom	False	False	0.0	0.00
00:10	Kitchen	False	False	0.0	0.00
00:10	Bathroom	False	False	0.0	0.00

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00:10	Home Office	False	False	0.0	0.00
00:15	Living Room	False	False	0.0	0.00
00:15	Bedroom	False	False	0.0	0.00
00:15	Kitchen	False	False	0.0	0.00
00:15	Bathroom	False	False	0.0	0.00
00:15	Home Office	False	False	0.0	0.00
00:20	Living Room	False	False	0.0	0.00
00:20	Bedroom	False	False	0.0	0.00
00:20	Kitchen	False	False	0.0	0.00
00:20	Bathroom	False	False	0.0	0.00
00:20	Home Office	False	False	0.0	0.00
00:25	Living Room	False	False	0.0	0.00
00:25	Bedroom	False	False	0.0	0.00
00:25	Kitchen	False	False	0.0	0.00
00:25	Bathroom	False	False	0.0	0.00
00:25	Home Office	False	False	0.0	0.00
00:30	Living Room	False	False	0.0	0.00

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00:30	Bedroom	False	False	0.0	0.00
00:30	Kitchen	False	False	0.0	0.00
00:30	Bathroom	False	False	0.0	0.00
00:30	Home Office	False	False	0.0	0.00
00:35	Living Room	False	False	0.0	0.00
00:35	Bedroom	False	False	0.0	0.00
00:35	Kitchen	False	False	0.0	0.00
00:35	Bathroom	False	False	0.0	0.00
00:35	Home Office	False	False	0.0	0.00
00:40	Living Room	False	False	0.0	0.00
00:40	Bedroom	False	False	0.0	0.00
00:40	Kitchen	False	False	0.0	0.00
00:40	Bathroom	False	False	0.0	0.00
00:40	Home Office	False	False	0.0	0.00
00:45	Living Room	False	False	0.0	0.00
00:45	Bedroom	False	False	0.0	0.00
00:45	Kitchen	False	False	0.0	0.00
00:45	Bathroom	False	False	0.0	0.00
00:45	Home	False	False	0.0	0.00

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	Office				
00:50	Living Room	False	False	0.0	0.00
00:50	Bedroom	False	False	0.0	0.00
00:50	Kitchen	False	False	0.0	0.00
00:50	Bathroom	False	False	0.0	0.00
00:50	Home Office	False	False	0.0	0.00
00:55	Living Room	False	False	0.0	0.00
00:55	Bedroom	False	False	0.0	0.00
00:55	Kitchen	False	False	0.0	0.00
00:55	Bathroom	False	False	0.0	0.00
00:55	Home Office	False	False	0.0	0.00
01:00	Living Room	False	False	0.0	0.00
01:00	Bedroom	False	False	0.0	0.00
01:00	Kitchen	False	False	0.0	0.00
01:00	Bathroom	False	False	0.0	0.00
01:00	Home Office	False	False	0.0	0.00
01:04	Living Room	False	False	0.0	0.00
01:04	Bedroom	False	False	0.0	0.00
01:04	Kitchen	False	False	0.0	0.00

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01:04	Bathroom	False	False	0.0	0.00
01:04	Home Office	False	False	0.0	0.00
01:10	Living Room	False	False	0.0	0.00
01:10	Bedroom	False	False	0.0	0.00
01:10	Kitchen	False	False	0.0	0.00
01:10	Bathroom	False	False	0.0	0.00
01:10	Home Office	False	False	0.0	0.00
01:15	Living Room	False	False	0.0	0.00
01:15	Bedroom	False	False	0.0	0.00
01:15	Kitchen	False	False	0.0	0.00
01:15	Bathroom	False	False	0.0	0.00
01:15	Home Office	False	False	0.0	0.00
01:19	Living Room	False	False	0.0	0.00
01:19	Bedroom	False	False	0.0	0.00
01:19	Kitchen	False	False	0.0	0.00
01:19	Bathroom	False	False	0.0	0.00
01:19	Home Office	False	False	0.0	0.00
01:25	Living	False	False	0.0	0.00

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	Room				
01:25	Bedroom	False	False	0.0	0.00
01:25	Kitchen	False	False	0.0	0.00
01:25	Bathroom	False	False	0.0	0.00
01:25	Home Office	False	False	0.0	0.00
01:30	Living Room	False	False	0.0	0.00
01:30	Bedroom	False	False	0.0	0.00
01:30	Kitchen	False	False	0.0	0.00
01:30	Bathroom	False	False	0.0	0.00
01:30	Home Office	False	False	0.0	0.00
01:34	Living Room	False	False	0.0	0.00
01:34	Bedroom	False	False	0.0	0.00
01:34	Kitchen	False	False	0.0	0.00
01:34	Bathroom	False	False	0.0	0.00
01:34	Home Office	False	False	0.0	0.00

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CHAPTER 10

APPENDIX C: EXTENDED LITERATURE REVIEW AND METHODOLOGY

Extended discussion on IoT protocols and standards 1. The Internet of Things (IoT) has revolutionized the way devices communicate within a smart home ecosystem. Protocols such as Zigbee, Z-Wave, and Bluetooth Low Energy (BLE) are commonly used for low-power, short-range communication between sensors and controllers. Wi-Fi and Ethernet provide higher bandwidth for connecting the IoT gateway to cloud services and user interfaces. MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol ideal for IoT applications, enabling efficient data transmission between devices with minimal network bandwidth. The integration of these protocols ensures seamless interoperability and robust performance in the Smart Home Lighting Automation System.

Detailed analysis of energy management algorithms 1. Intelligent energy management requires sophisticated algorithms to balance user comfort with energy efficiency. Rule-based systems, like the one implemented in this project, use predefined thresholds and logic to control devices. Advanced systems may incorporate fuzzy logic to handle the uncertainty and imprecision of sensor data, or machine learning models to adapt to user preferences over time. Predictive control algorithms can anticipate occupancy based on historical patterns and adjust lighting proactively. The continuous optimization of these algorithms is crucial for maximizing energy savings and enhancing the overall smart home experience.

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